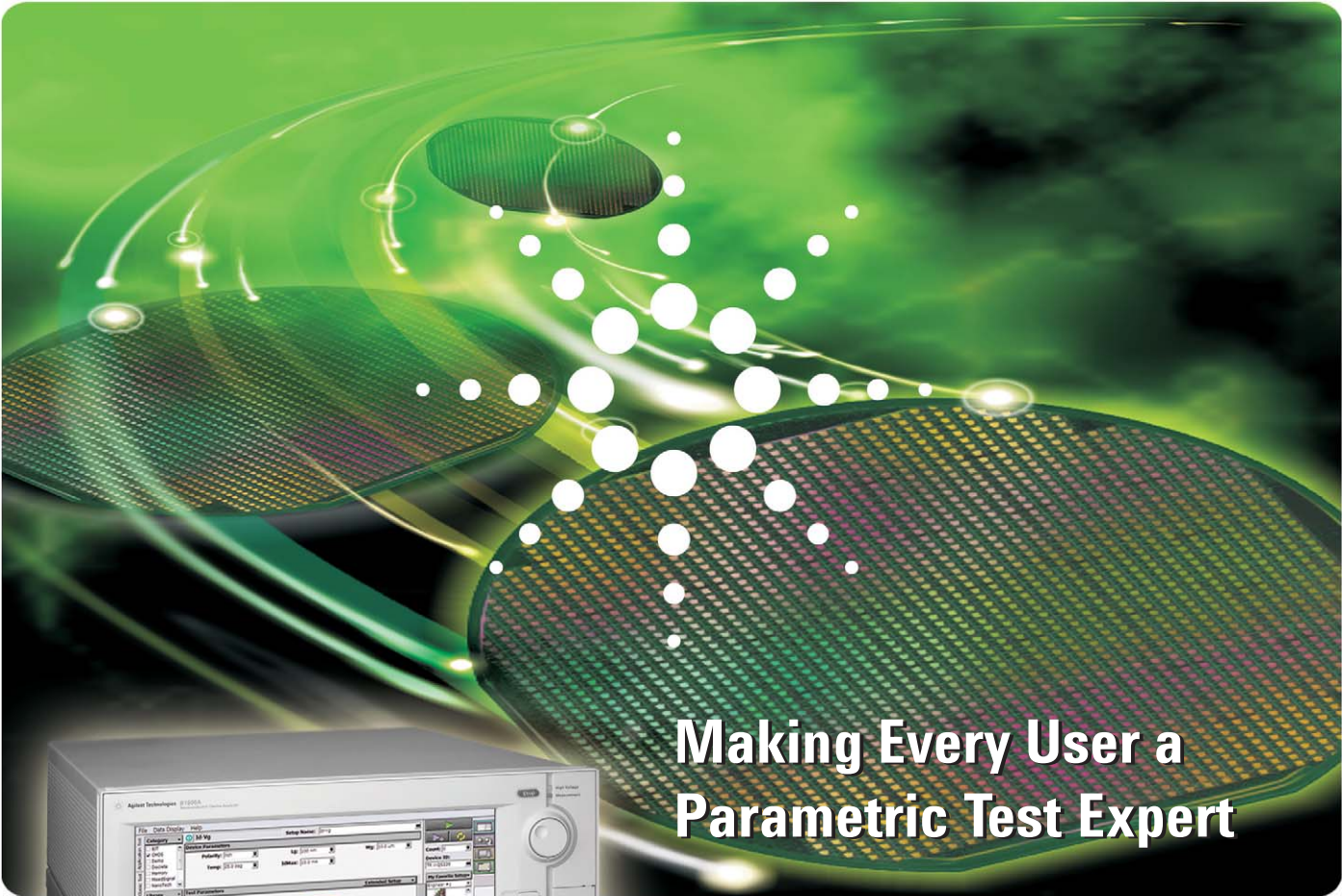
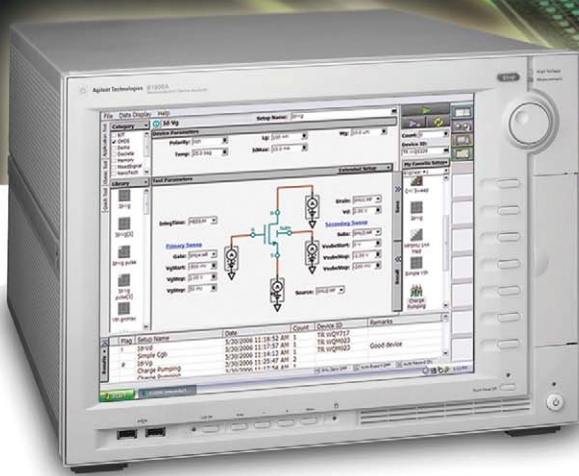




Making Every User a Parametric Test Expert



**Agilent B1500A Semiconductor Device
Analyzer with EasyEXPERT Software**

Agilent Desktop EasyEXPERT Software



Agilent Technologies

The total parametric analysis environment...

The image shows the Agilent B1500A Semiconductor Device Analyzer with the EasyEXPERT software interface. The interface includes a menu bar (File, Data, Display, Help), a toolbar, and several windows. The main window displays a circuit diagram for an Id-Vg test, with various parameters like Polarity, Lgt, Wgr, Temp, IdMax, and IntegTime. A 'My Favorite Setup' window on the right shows a list of saved setups. A 'Results' window at the bottom displays a table of test results.

Callouts and features highlighted:

- Microsoft Windows-based EasyEXPERT software**: Points to the software interface on the device.
- More than 150 furnished application tests**: Points to the 'Library' window showing various test templates.
- Innovative, task-based approach to parametric test**: Points to the main test setup window.
- Measure, append and repeat functions**: Points to the 'My Favorite Setup' window.
- Built-in, semiautomatic wafer probe drivers**: Points to the physical probe drivers on the right side of the device.
- Integrated switching matrix control for the B2200A and B2201A**: Points to the switching matrix control area.
- "Quick Test" utility supports test sequencing without programming**: Points to the 'Quick Test' utility in the toolbar.
- Intuitive, GUI-based application test setup window**: Points to the main test setup window.
- "My Favorite Setup" feature for customizing application tests**: Points to the 'My Favorite Setup' window.
- Automatically export data during measurement**: Points to the 'Auto Export' checkbox in the bottom status bar.
- Data and setup information automatically saved after each measurement**: Points to the 'Save' button in the bottom status bar.

The best choice for all your parametric characterization needs

The Agilent B1500A semiconductor device analyzer with EasyEXPERT software is a complete parametric test solution. It supports all aspects of parametric test, from basic manual measurements to test automation across a wafer in conjunction with a semiautomatic wafer prober. Because the B1500A utilizes the Microsoft® Windows® XP Professional operating system, it integrates easily into your PC-based work environment. Best of all, the familiar Windows graphical user interface (GUI) and convenient online help menus minimize the need for instrument training.

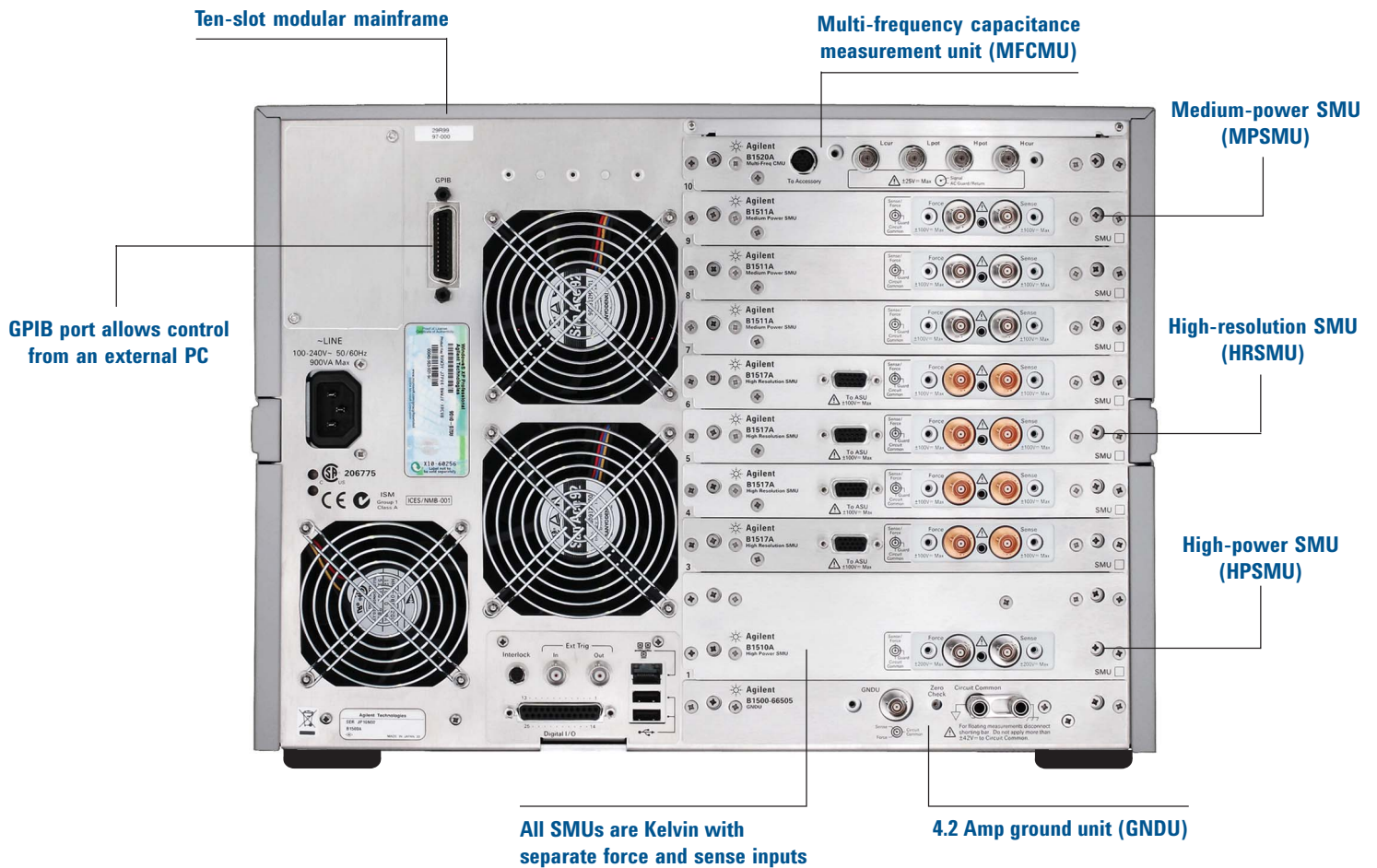
Desktop EasyEXPERT improves productivity

Desktop EasyEXPERT software provides all of the benefits of the B1500A EasyEXPERT interface on a stand-alone PC. In offline mode you can perform important tasks such as analyzing data and creating new application tests, which frees up the B1500A to function in its primary role of making measurements. In online mode you can control the B1500A (as well as other instruments) via GPIB from a separate PC, which eliminates the need to interface directly with the B1500A.

An adaptable, scalable solution

The ability to run EasyEXPERT either from the B1500A or from a standalone PC gives you great flexibility. Now you can work just as effectively at your desk as you can in your lab. It is easy to start with a B1500A and gradually build up a racked, integrated solution containing switching matrices and other instruments. This ability removes the limitations of traditional instrument-based parametric characterization and maximizes your efficiency.

...that reduces cost and complexity



Modular design minimizes costs and protects your investment

With ten module slots and a wide selection of source/monitor unit (SMU) types, you can configure the B1500A exactly the way you want. You also have ample room for expansion if your measurement needs change in the future. And, as Agilent makes new modules available, you can easily add new test capabilities to your B1500A. This ability to meet both known and unanticipated test needs ensures that your parametric test investment is protected into the future.

Integrated capacitance module improves CV measurement

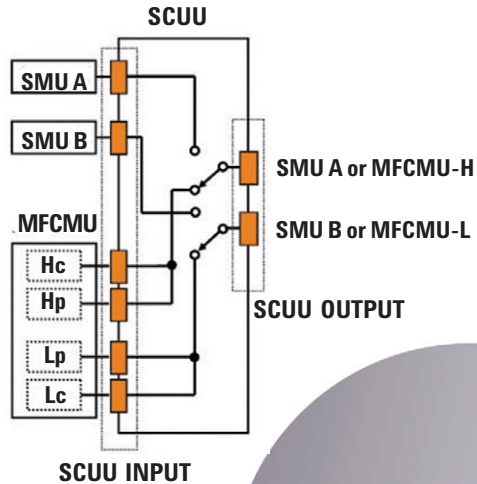
The B1500A's multifrequency capacitance measurement unit (MFCMU) eliminates the need for a separate external capacitance meter. In addition, an optional SMU CMU unify unit (SCUU) supports capacitance versus voltage (CV) and current versus voltage (IV) switching in positioner-based wafer probing environments. This completely integrated solution eliminates the common measurement errors that frequently occur when using a separate external capacitance

meter. The tight integration of hardware and software makes switching between CV and IV measurements effortless, while providing unprecedented accuracy. In addition, the ability to automatically stabilize the series inductance in the CV measurement path provides dramatic improvement in repeatability. Finally, this solution saves bench space while reducing cost and complexity.

SCUU and ASU solve positioner-based switching challenges

A CV-IV solution for every need

The B1500A supports two positioner-based switching solutions. Depending upon which SMU resources you choose, Agilent can supply solutions with 0.5 μV voltage measurement resolution and 10 fA, 1 fA, or 0.1 fA current measurement resolution.



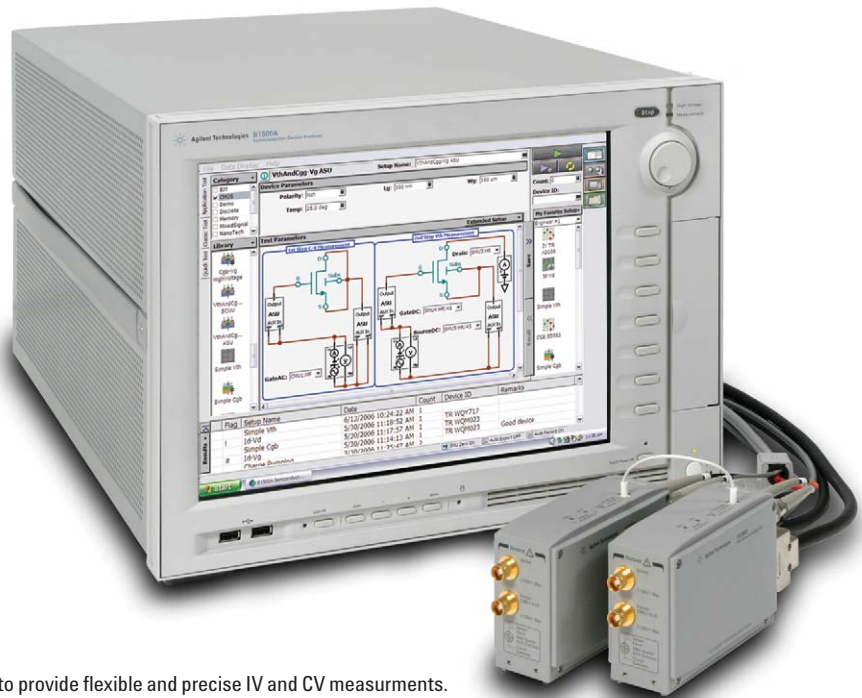
SCUU makes accurate CV-IV measurements fast and effortless

The SMU CMU unify unit (SCUU) works with the B1500A's MFCMU and MPSMUs or HRSMUs. The SCUU enables you to switch between CV and IV measurements without sacrificing any measurement accuracy. Besides taking care of switching and accuracy issues the B1500A and SCUU combination also solves CV measurement compensation, connection and return path issues. All of this is done automatically, without requiring you to have a detailed understanding of the theory involved in making CV measurements.



ASU provides 0.1 fA measurement with CV-IV switching

The B1500A HRSMU accepts an optional attosense and switch unit (ASU), which increases the low-current measurement resolution of the HRSMU to 100 attoamps (0.1 fA). The ASU also has two BNC inputs that can work with the B1500A MFCMU. Using two ASUs, you can switch easily between ultra-low current and voltage (IV) measurements and capacitance (CV) measurements without having to change any cables.

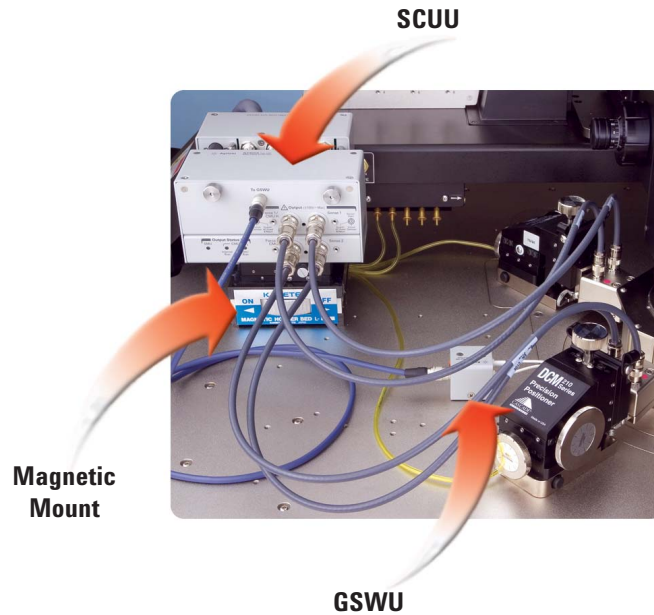


The ASU works with the HRSMUs and the MFCMU to provide flexible and precise IV and CV measurements.

Multiple positioner-based CV-IV wafer probing solutions

Easy integration into wafer probing environments

A cable extension and magnetic mount are available to allow placement of the SCUU on your probe in close proximity to the wafer. You can also use the optional Guard Switch Unit (GSWU) in conjunction with the SCUU and your wafer probe positioners to ensure that a current return path is available during capacitance measurement. These capabilities combine to give you unprecedented CV measurement accuracy and repeatability.



10 fA/0.5 μ V Solution

1 x MFCMU
2 x MPSMU
1 x SCUU

1 fA/0.5 μ V Solution

1 x MFCMU
2 x HRSMU
1 x SCUU

0.1 fA/0.5 μ V Solution

1 x MFCMU
2 x HRSMU
2 x ASU

Kelvin Measurement
Medium Frequency Capacitance Measurement
CV-IV Switching
Integrated Capacitance Compensation

You can combine the B1500A's MFCMU and MPSMUs or HRSMUs with the SCUU or ASUs to create positioner-based CV-IV switching solutions to meet a variety of needs. All solutions are Kelvin, they support CV-IV switching, and they have integrated capacitance compensation.

ASU direct mounting option minimizes cable length

The ASU can also be mounted directly on the wafer probe to minimize excess cable length. Some semi-automatic wafer probe companies even support directly mounting the ASUs onto the wafer probe positioners. This maintains the 0.1 fA measurement resolution capability of the ASU and reduces the need to compensate for the additional cable length going to the probe tips.



Quick startup with support for multi-user environments

EasyEXPERT makes parametric test as easy as 1-2-3.

1 Select one or more technology categories

2 Select desired application test from the library

3 Click or press the Measure button

A graphical plot and data list are generated automatically in real time.

Flag	Setup Name	Date	Count	Device ID	Remarks
!	Id-Vd	5/30/2006 11:18:52 AM	1	TR WQY717	
#	Simple Cgb	5/30/2006 11:17:57 AM	1	TR WQM023	
	Id-Vg	5/30/2006 11:14:13 AM	1	TR WQM023	Good de
	Charge Pumping	3/30/2006 11:25:47 AM	2		
	Charge Discharge	3/30/2006 11:17:54 AM	1		

Revolutionary task-oriented approach to parametric test

EasyEXPERT represents a paradigm shift from previous and competing parametric instruments. It allows you to focus on your real goal of making efficient parametric measure-

ments, without becoming an instrument hardware expert. EasyEXPERT comes with more than 150 application tests conveniently organized by device type, application, and technology. In addition, you can easily modify and customize the furn-

ished application tests to fit your specific needs. Instead of spending hours or days learning how to set up the instrument hardware, you can begin making productive parametric measurements immediately.

My Favorite Setup

Engineer #1

- C-V Sweep
- Id-Vg
- MPSMU 1nA Med
- Simple Vth
- Charge Pumping

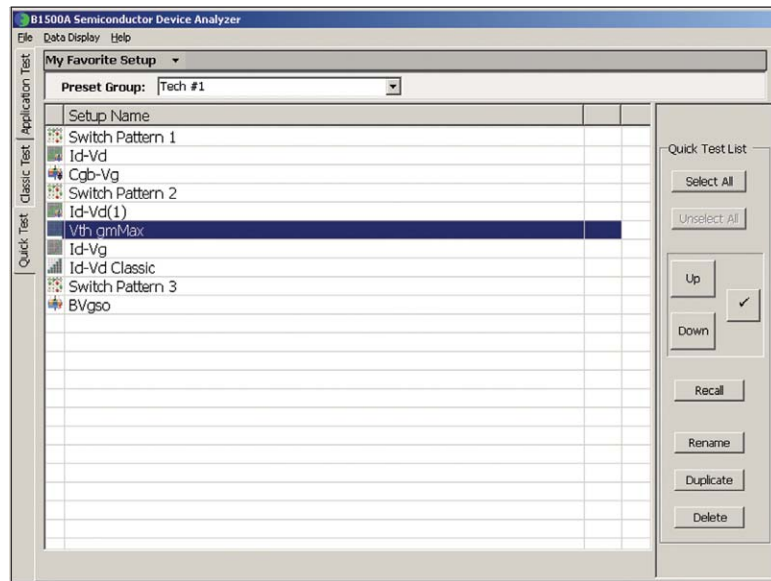
Test the way you want every time you use the B1500A

EasyEXPERT supports a "My Favorite Setup" feature that enables you to create, save, and recall your own set of unique, customized application tests. These tests are saved directly to the hard drive, so they are readily available when needed. There is no need to deal with floppy disks or other PC media. The ability to save customized versions of application tests to a personal list of favorites permits even occasional B1500A users to become immediately productive each time they use the instrument.

Effortless test sequencing and programming

Quick Test allows fast and easy test sequencing

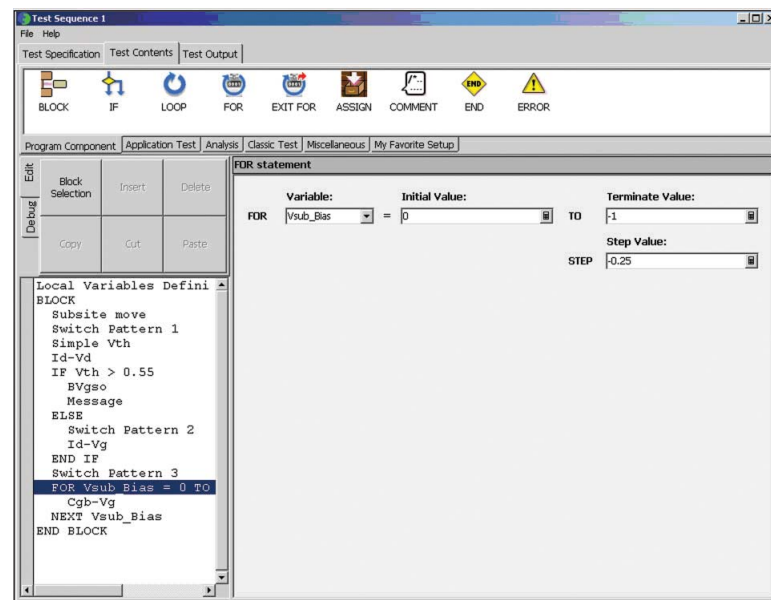
EasyEXPERT has a GUI-based Quick Test mode that enables you to perform test sequencing without programming. By simply selecting an existing “My Favorite Setup,” you can then select, copy, rearrange and cut-and-paste any application tests with a few simple mouse clicks. Once you have selected and arranged your tests, simply click on the measurement button to begin running an automated test sequence.



Quick Test enables you to perform test sequencing without having to do any programming.

Application test editor supports powerful programming features

The intuitive GUI-based application test editor included with EasyEXPERT does not require you to become familiar with a complicated programming language to modify or create application tests. The test editor supports all standard programming operations, such as performing conditional branching (If/Then/Else, For/Next, Loop/Until, etc.), controlling other instruments and even calling other application tests. This makes it easy to create sophisticated testing algorithms that are both modular and flexible. And since all measurement resources are treated as variables, it is easy to modify pin assignments and cascade these changes down to all of the application tests that you are calling.



The Application Test editor allows you to create and modify application tests without having to learn a complicated programming language. You can also easily string together sequences of other application tests.

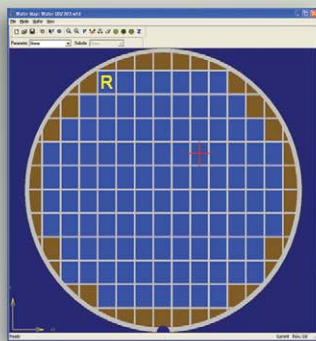
EasyEXPERT and Desktop EasyEXPERT in the Lab . . .



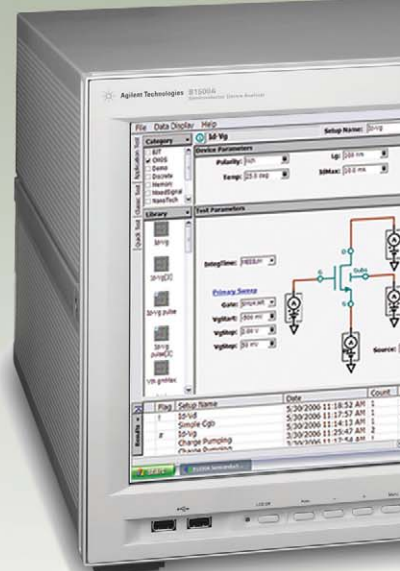
Switching Matrix Control



CV-IV and Pulsed I-V



Wafer Prober Control



ONLINE IN THE LAB — EasyEXPERT software running on the B1500A or Desktop EasyEXPERT running on an external PC lets you control all measurements to test automation across a wafer in conjunction with a semiautomatic wafer prober. It is easy to start with a B1500A and gradually build out your system with switching matrices and other instruments. This solution removes the limitations of traditional instrument-based parametric characterization and measurement.

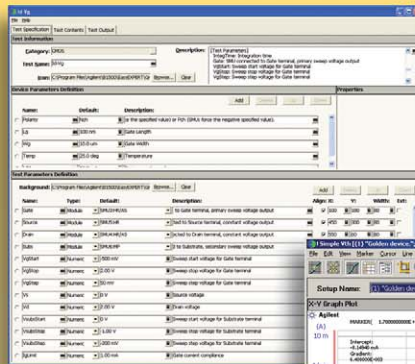
...Desktop EasyEXPERT in the Office



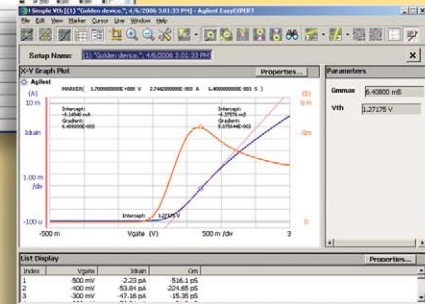
IV Measurements



Automatic
Data Export



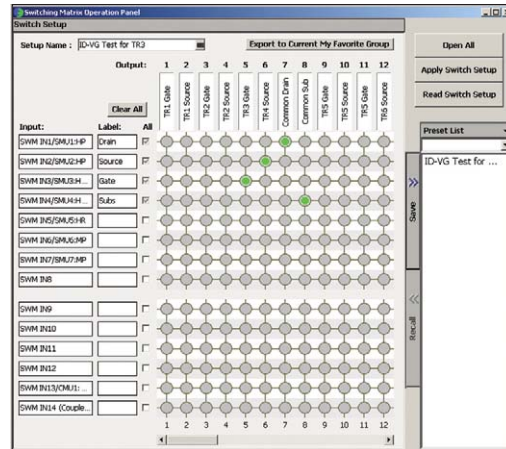
Data Analysis and
Application Test
Development



aspects of parametric test, from basic manual
build up a racked, integrated solution containing
maximizes your efficiency.

OFFLINE IN THE OFFICE — Desktop EasyEXPERT software running on a stand-alone PC provides all the benefits of the B1500A EasyEXPERT interface. In offline mode you can perform tasks such as analyzing data and creating new application tests.

Fully integrated switching matrix control



The B2200A and B2201A switching matrices have 8 triaxial inputs and 6 BNC inputs (14 internal paths), and they support up to 48 triaxial outputs. EasyEXPERT provides an intuitive GUI to control the B2200A and B2201A switching matrices.

Superior switching matrix performance and design

EasyEXPERT has built-in drivers to control the Agilent B2200A femtoamp leakage switch and the Agilent B2201A 14-channel low-leakage switch. These switching matrices both provide exceptional low-current leakage performance. The B2200A and B2201A allow you to achieve

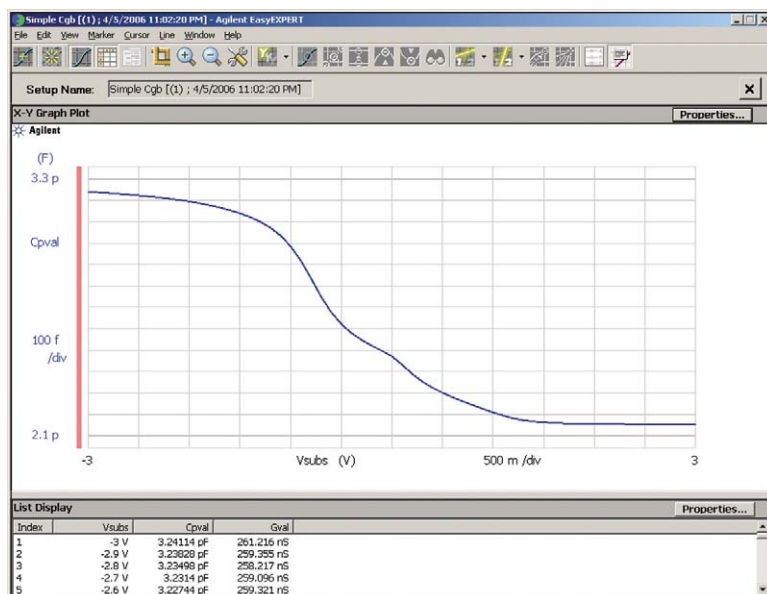
measurement resolutions of 1 fA, and 10 fA respectively. This enables you to realize the full measurement potential of your B1500A while benefiting from the measurement flexibility inherent in a switching matrix based solution.

Intuitive GUI switching matrix control

EasyEXPERT allows you to create and save switch setups via a Windows-based GUI. You can connect or disconnect any input and output by simply clicking on an intersection point. You can also create mnemonic labels for the output ports. Once saved, switching matrix setups can be recalled programmatically from either Quick Test or an application test.

Intrinsic capacitance compensation

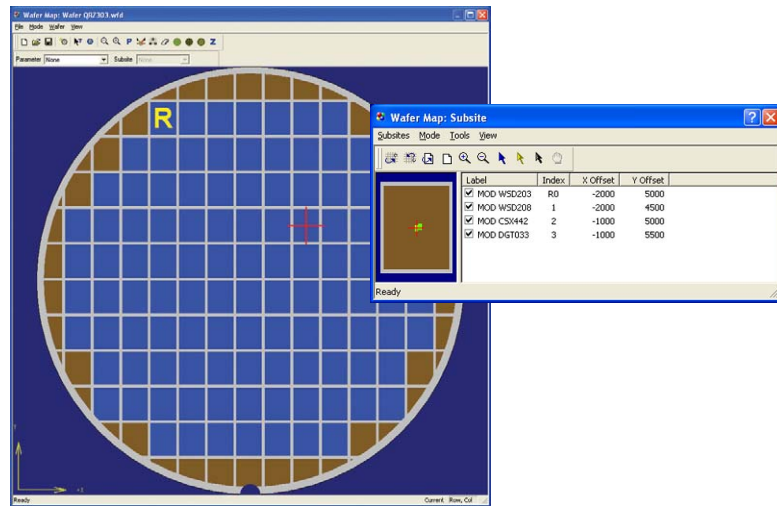
Switching matrix capacitance compensation is supported in EasyEXPERT. Cable lengths are automatically compensated up to the output of the switching matrix, and you can create your own compensation file to correct for the additional cables and probes going out to the device under test (DUT) if desired.



Powerful and intuitive automated test

Complete wafer test automation

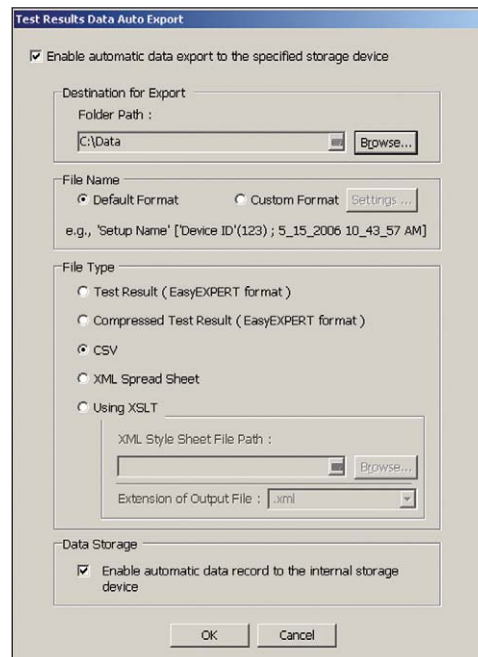
All popular semiautomatic wafer probers are supported by EasyEXPERT. This makes it easy for you to define the wafer, die, and module information for probing across an entire wafer. You can combine wafer prober control with either Quick Test mode or an application test-based test sequence to perform multiple testing on various devices across the wafer. If you are using a probe card and switching matrix, you can also call switching patterns automatically.



Wafer prober drivers are included for all of the popular semiautomatic wafer probers, and additional drivers will be added over time.

Automatic data export

EasyEXPERT has the ability to automatically export measurement data in real time, in a variety of formats. You can save data to any drive connected to the B1500A or, in the case of Desktop EasyEXPERT, any drive connected to a PC. If you wish, you can export data to a network drive and view test results on your desktop PC as your instruments are performing the testing in your lab.



The auto data export feature allows you to export your measurement data to any accessible location (network drive, USB memory device, etc.) in real time as you are making measurements.

Conveniently control other instruments

The EasyEXPERT application test editor allows you to send and receive GPIB commands. This makes it easy to control other instruments in your lab. It is a simple task to automate this feature and use it to perform operations such as turning on a thermal unit in coordination with other automated testing activities.

Pulsed IV measurement down to 10 nanoseconds

Modern devices present new measurement challenges

Silicon-on-insulator (SOI) transistors and high-k gate dielectric transistors present significant new measurement challenges. SOI transistors are extremely sensitive to thermal effects, so they require extremely short duty cycles during characterization. High-k gate dielectric transistors are subject to electron tunneling and charge trapping effects, which also require ultra-short pulse widths for accurate characterization.

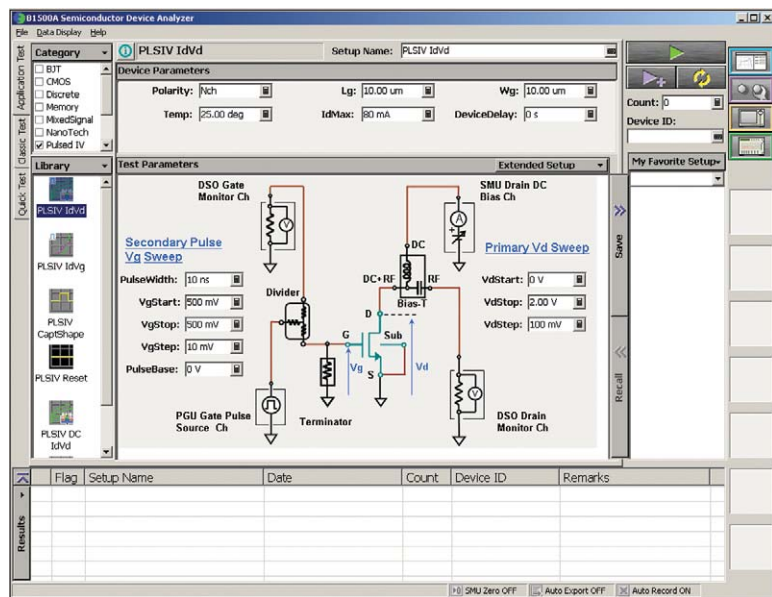
Pulsed IV solution from lab to fab

A pulsed IV option is available for the B1500A that supports pulsed IV measurement down to 10 ns (with 2 ns rise and fall times). The solution consists of the B1500A with a pulsed IV application test library, an Agilent Digital Storage Oscilloscope, Agilent Pulse Generator, cables and dc/RF connector modules. This single-vendor solution is completely self-contained and ready to use. You need only to connect the equipment, load the application test, and click the Measure button. An optional switch module is also available to enable you to toggle between dc and pulsed measurements without having to change any cables.

If you already own a compatible Agilent oscilloscope and pulse generator, you can purchase only the software, cables, and connectors. This well-proven solution is also available as an option for the Agilent 4075 and Agilent 4076 production test systems. This option allows you to correlate pulsed IV measurements in your lab with measurements made in production.



Agilent's pulsed IV solution uses well-proven Agilent instrumentation to provide a complete single-vendor solution to characterize SOI and high-k gate dielectric transistors.

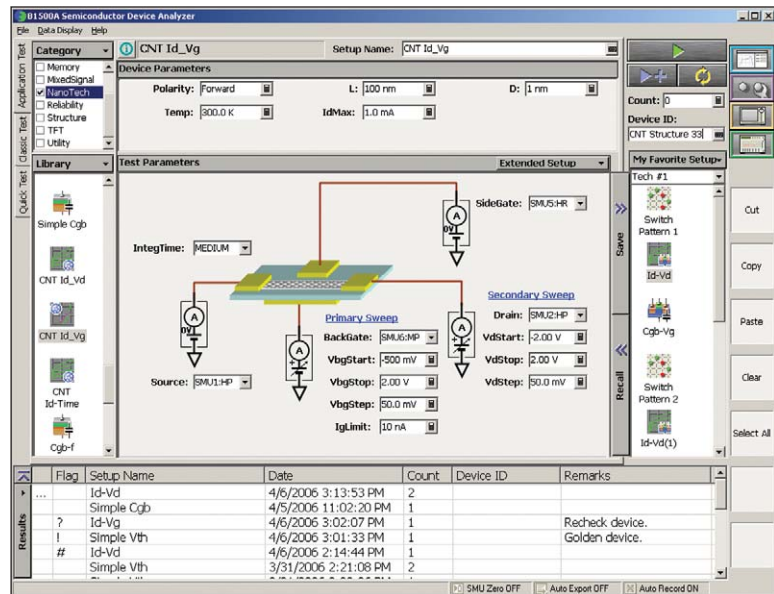


EasyExpert provides an intuitive user interface for the Pulsed IV application.

Measurement solutions at the nanoscale level

Nanodevice characterization

The B1500A comes with numerous application tests specifically designed for the dynamic and rapidly evolving field of nanodevice electrical characterization. For nanodevice researchers not familiar with all of the nuances of electrical characterization, this application test library enables the rapid and accurate measurement of structures such as carbon nanotube (CNT) field effect transistors (FETs) without having to spend valuable time learning how to set up the hardware.

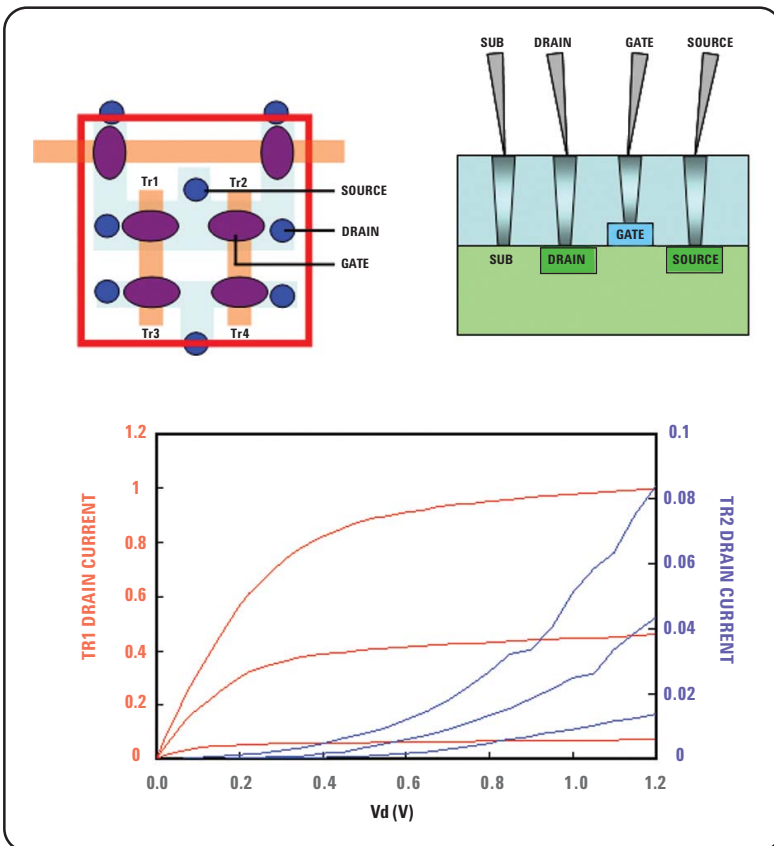


You can easily characterize exotic structures such as carbon nanotube (CNT) field effect transistors (FETs) using the B1500A.

Nanoprobng and failure analysis

The B1500A's EasyEXPERT software has many features that make it an excellent tool for performing failure analysis on nanoscale structures. In the past, for example, it has been difficult to determine which transistor in a failing memory cell is defective. Since EasyEXPERT has the ability to dynamically reconfigure SMU assignments and automatically run multiple tests in sequence, it is straightforward to characterize each transistor in the cell until the failing device is identified.

Agilent is working with scanning electron microscope (SEM)-based and atomic force probe (AFP)-based prober suppliers to provide comprehensive nanoprobng solutions. This enables you to probe extremely tiny traces on-wafer such as the first metal layer (M1), and to electrically evaluate these structures using the B1500A.

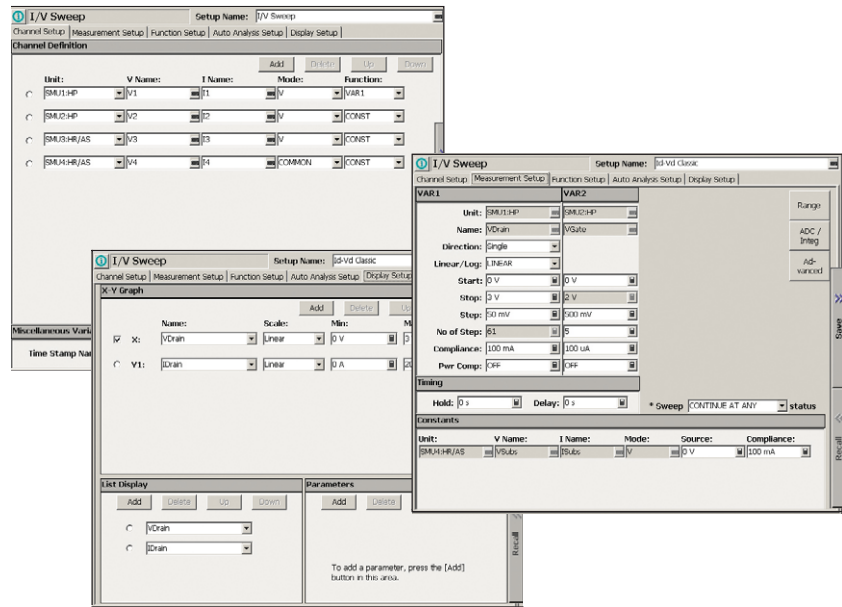


The plots above show how you can use the B1500A and EasyEXPERT software to sequentially evaluate transistors in a memory cell until the failing device is identified.

Convenient migration paths for Agilent 4155 and 4156 users

Classic Test mode offers a familiar yet improved user interface

For users who prefer the 4155/4156 user interface, the B1500A offers a classic test mode. This mode not only maintains the look, feel, and terminology of the 4155/4156 interface, it enhances user interaction by taking full advantage of Microsoft Windows GUI features. If you are an experienced 4155 or 4156 user, you can enjoy all of the benefits of the B1500A hardware and software, without having to change your work preferences.



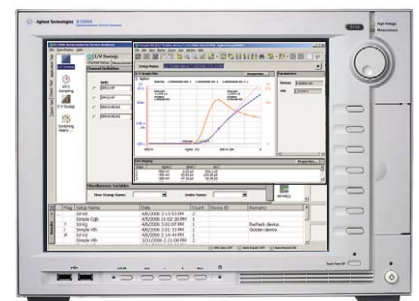
If you are an experienced 4155 or 4156 user, you can use the Classic Mode interface while still taking advantage of the new B1500A hardware and software features.

Agilent 4155/4156 setup converter provides easy transition to the B1500A

Agilent supplies a free setup converter tool that runs on any Windows-based PC. This enables effortless transfer of existing 4155/4156 measurement setup files to the B1500A. The tool takes 4155 and 4156 measurement setup files (.MES) and converts them into equivalent EasyEXPERT 4155/4156 classic test mode setup files. By eliminating the need to recreate measurement setups, you can accelerate your transition to the B1500A.

Convert 4155 or 4156 programs to control the B1500A

An available *VXIplug&play* driver minimizes work when converting 4155/4156 SCPI-based programs to the B1500A. If you have 4155 or 4156 programs that use Agilent FLEX commands, you can transport your program to control the B1500A with only minor modifications. Application assistance is also available to help you with the task of converting your old programs.



The 4155/4156 setup converter tool makes it easy to use your existing 4155 and 4156 measurement setup files on the B1500A.

Mainframe characteristics and module selection guide

Mainframe Characteristics

Available slots	10
Ground unit sink capability	4.2 A
USB ports	2 front and 2 rear
Instrument control	GPIB
Networking	100BASE-TX/10BASE-T LAN Port
External trigger inputs/outputs	1 BNC Trigger In; 1 BNC Trigger Out; 8 Programmable Trigger In/Out

Module Selection Guide

Module	HPSMU (B1510A)	MPSMU (B1511A)	HRSMU (B1517A)	ASU (E5288A)	MFCMU (B1520A)
Required slots	2	1	1	NA	1
Maximum force voltage	± 200 V	± 100 V	± 100 V	± 100 V	NA
Maximum force current	± 1 A	± 100 mA	± 100 mA	± 100 mA	NA
Voltage measurement resolution	2 µV	0.5 µV	0.5 µV	0.5 µV	NA
Current measurement resolution	10 fA	10 fA	1 fA	0.1 fA	NA
Minimum/maximum capacitance measurement frequency	NA	NA	NA	NA	1 kHz to 5 MHz
Maximum capacitance measurement dc bias capability	NA	NA	NA	NA	± 100 V*

*When used in combination with the SCUU and SMUs

EasyEXPERT software characteristics

Data Acquisition and Control

User interface	Windows GUI
User interface options	Touchscreen, softkeys, clickable knob, USB keyboard and mouse
Data viewing	View multiple graphs in tile, tab, or overlay display formats
Data storage	Data and test settings automatically saved after each measurement
Test sequencing	Test sequencing without programming via Quick Test mode
Switching matrix control	Integrated switching matrix control for Agilent B2200A/B2201A
Wafer prober control	Integrated semiautomatic wafer prober control supports die/sub-die moves
Supported wafer probers	Cascade Microtech, Micromanipulator, SUSS MicroTec, Vector Semiconductor
CV measurement	Integrated capacitance compensation
Application test support	More than 150 furnished application tests conveniently organized by technology
Application test management	Workspace feature allows creation of public/private application test libraries
Alternative user interface	Agilent 4155/4156 Classic Test mode

Plotting and Reporting

Data analysis	Automated real-time graphical data analysis
Data comparison	Append feature displays multiple measurements on same graph
Data export	Automatic data export to any available drive
Printing	Print to any Windows-supported printer (via LAN or USB)

Environment and Connectivity

Operating system	Windows XP Professional
Hardcopy media	DVD-ROM / CD-ROM / CD-RW Drive
Networking	Windows XP Professional
Supported peripherals	Any PC peripheral using USB

Web Resources

B1500A Web Site

Visit our B1500A Web site for additional product information and literature.

www.agilent.com/see/B1500A

Agilent E-mail Updates

Get the latest information on the products and applications you select.

www.agilent.com/find/emailupdates

Key B1500A Accessories

B1510A	High-Power Source/Monitor Unit
B1511A	Medium-Power Source/Monitor Unit
B1517A	High-Resolution Source/Monitor Unit
B1520A	Multifrequency Capacitance Measurement Unit
E5288A	Attosense and Switch Unit
N1254A-100	Ground Unit-to-Kelvin Adapter
N1300-001/002	Capacitance Measurement Unit Cable (1.5 meter/3 meter)
N1301A-100	SMU CMU Unify Unit
N1301A-102	SCUU Cable (3 meter)
N1301A-200	Guard Switch Unit
16444A-001	USB Keyboard
16444A-002	USB Mouse
16444A-003	Stylus
16442B	Test Fixture
16494A-001/002	Triaxial cable (1.5 meter/3 meter)



B1500A with optional keyboard and mouse.

For more information about Agilent and its products, go to www.agilent.com.

For more information about the B1500A Semiconductor Device Analyzer and EasyEXPERT software, please call one of the centers listed below and ask to speak with an Agilent sales representative.

Americas

Brazil (11) 4197-3600
Canada (French) 1 877 894-4414
Canada (English) 1 800 447-8378
Mexico 33 134-5841
United States 1 800 829-4444

Asia/Asia Pacific

Australia 1 800 629-485
China 1 800 276-3059
Hong Kong 852 2599 7889
India 91/11 690-6156
Japan 0120 421-345
Malaysia 1 800 880-780
New Zealand 0 800 738 378
Philippines 1 800 1651-0135
Singapore 1 800 276-3059
South Korea 080 778-0011
Taiwan 0 800 047-662
Thailand 1 800 2758-5822

Europe

Austria (01) 25 125-7183
Belgium (0) 2 404-9380
Denmark 080301040
Finland 20 547-9999
France (0) 825 010710
Germany (0) 18 05 24-63 34
Greece 20 547-9999
Ireland 016158393
Italy 02 92 60 8333
Luxembourg (0) 2 404-9340
Netherlands (0) 20 547-9999
Poland 20 547-9999
Russia 20 547-9999
Spain 91 631 3383
Sweden 020 120-9975
Switzerland (Italian) (0) 2 92 60 8484
Switzerland (German) (0) 1 735-9300
Switzerland (French) (0) 825 010 700
United Kingdom (0) 7004 222-222

Middle East

Israel 20 547-9999

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